MALLA REDDY ENGINEERING COLLEGE FOR WOMEN
(Autonomous Institution-UGC, Govt. of India)
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Maisammaguda, Dhulapally, Secunderabad 500100.

SMART GARDENING AND SAFETY MONITORING SYSTEM

ABSTRACT:

This project presents a smart gardening system utilizing an Arduino Uno, Bluetooth communication, and various sensors to automate and monitor plant health. The system employs a soil moisture sensor to determine soil dryness. When the soil moisture level falls below a predefined threshold, a relay activates a water pump to irrigate the plants. Conversely, if the soil is sufficiently moist, the pump remains off. A fire sensor is integrated to detect potential fire hazards, triggering a buzzer alarm upon detection. Real-time data from the sensors, including soil moisture and fire detection status, is transmitted via Bluetooth to a connected device, enabling users to remotely monitor and control the gardening environment. This system provides an efficient and automated solution for maintaining optimal plant growing conditions while enhancing safety

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